

Hyeongseob Jo

Seoul, South Korea
johyeongseob725@gmail.com
[GitHub](#) | [LinkedIn](#) | [Blog](#)

BIO

Machine Learning Engineer at ML2 specializing in on-device vision-language model deployment and optimization. Graduate research focused on computer vision for industrial defect classification.

FIELD OF INTEREST

Computer Vision, Defect Classification, Representation Learning, Multimodal Learning

EDUCATION

- Dongguk University**
M.S. in Computer Science and Artificial Intelligence

Feb. 2026
- Dongguk University**
Bachelor of Science in Engineering in Mechanical Engineering

Feb. 2024
- Andong National University**
Bachelor of Arts in Ethics Education; Double Major in English Education

Feb. 2020

POSITIONS

- Machine Learning Engineer**
[ML2](#)
Optimized and quantized MobileVLM for on-device iOS deployment using ONNX and Core ML.
Collaborated in weekly technical discussions in English.

Dec. 2025 – May. 2026
- Graduate Student Research Assistant**
[SVM Lab](#)
Conducted graduate-level research in computer vision.
Developed a multi-illumination defect classification algorithm for display panels.
Industry-academic cooperation project with LG Display (Mar. 2024 – Feb. 2025).

Mar. 2024 – Feb. 2026
- High School Teacher (Contract)**
[Youngju Girls' High School](#)
Taught Integrated Social Studies (Grade 1), Philosophy, Ethics (Grade 2)

Mar. 2020 – Dec. 2020

PROJECT

Google Summer of Code (Upcoming)

Summer 2026

Project: [Agentic AI for Predictive Maintenance using OpenVINO](#)

- Selected as a contributor for Google Summer of Code (GSoC) 2026.
- Program duration: May 25 – Aug. 24, 2026.

Samsung Collegiate Programming Challenge

Summer 2025

AI Competition: [Multimodal AI for Everyday Photo Understanding](#)

- Proposed CLIP · SigLIP-based ensemble model for multimodal image-text classification.
- Achieved 71% accuracy on test set (851 samples), ranked 41st out of 242 participants (Top 50 finalist).

LG Aimers

Winter 2023

AI Competition: [B2B Sales Opportunity Prediction](#)

- Developed a binary classification model to predict B2B sales opportunities from MQL data.
- Benchmarked 10+ ML models (e.g., XGBoost, LightGBM) for B2B sales opportunity prediction.

Autonomous Robot Navigation

Spring 2023

Dongguk University Course Project ([MEC4092-01](#))

- Implemented ROS2-based autonomous navigation with real-time control and sensor-fusion.
- Evaluated navigation performance in a physical testbed environment with obstacle avoidance tasks.

PUBLICATIONS

International Conferences

1. Hyeongseob Jo, Seunggi Park, and Sung In Cho. [A Survey of Unsupervised Learning-Based Out-of-Distribution Detection](#). In *Proceedings of the IEEE/IEIE International Conference on Consumer Electronics-Asia (ICCE-Asia)*, Nov. 2024
2. Y. S. Oh, M. G. Kim, B. Kim, J. Y. Kim, Hyeongseob Jo, J. H. Park, G. Lee, and S. I. Cho. [Dual-Branch Network via Multiple Illumination-Aware Representation Learning for Steel Surface Defect Classification](#). In *Proceedings of the British Machine Vision Conference (BMVC)*, Sheffield, UK, Nov. 2025
3. Yoonseo Park, Hyeongseob Jo, and Sung In Cho. [A Survey of Training-free Diffusion-based Image Generation with Free-form Mask](#). In *Proceedings of the 2025 International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC)*, Jul. 2025

Domestic Conference (Korean)

1. 이선호, 남혁, 조형섭, and 조성인. [차선 검출: 주요 접근법과 최신 연구 동향](#). In *대한전자공학회 하계학술대회 논문집*, pages 2969–2973, 2025

Magazine Article (Korean)

1. 조형섭, 오용석, and 조성인. [다중 조명 환경에서 제품 외관 검사 기법에 대한 이해](#). *Information Display (한국정보디스플레이학회)*, 26(6), Dec. 2025

Dissertations

1. 조형섭. [산업 공정의 다중 조명 환경에서 표면 결함 자동 검사를 위한 경량 딥러닝 기반 학습방법](#). Master's thesis, Dongguk University, Seoul, Korea, Feb 2026

PATENTS

Domestic Patent

[1] 조성인, 조형섭, 이미지 내 결함을 분류하는 시스템 및 분류 방법, 출원 10-2025-0155364 (2025-10-24).

TEACHING ASSISTANT

AI course (machine learning, deep learning)

Jun. 10–13 and 23–25 2025

LG Display

Introduction to Industry-Academia Project (CAI7177-01)

Spring 2025

Computer Science and Engineering, Dongguk University

Signal Processing, Image Processing

Aug. 19–23 2024

LX Semicon

Creative Engineering Design (CSC2004-03)

Spring 2024

Computer Science and Engineering, Dongguk University

LANGUAGES & SKILLS

- Korean (native), English (working proficiency)
- AI Frameworks: PyTorch, HuggingFace Transformers
- Deployment: ONNX Runtime, Core ML
- GPU Computing: CUDA
- Languages: Python, Swift, MATLAB
- OS: Linux (Ubuntu), macOS, Windows

Revised May 16, 2026